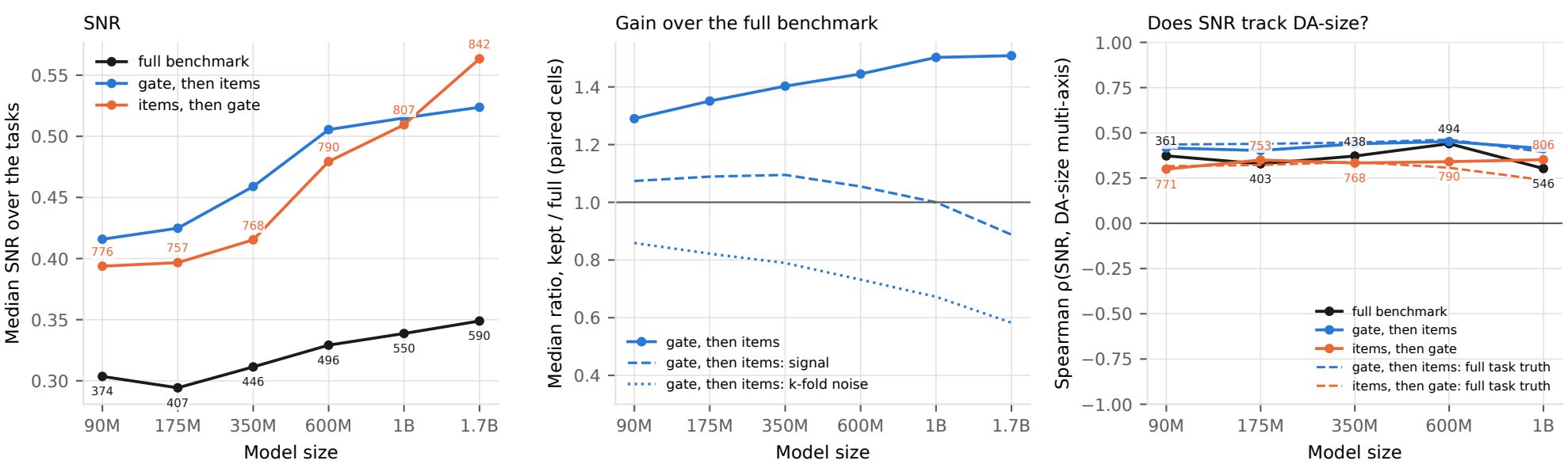
SNR
full benchmark
gate, then items
items, then gate
Median SNR over the tasks
0.55
0.50
0.45
0.40
0.35
0.30
842
807
790
768
776
757
590
550
496
446
374
407
90M
175M
350M
600M
1B
1.7B
Model size
Gain over the full benchmark
Median ratio, kept / full (paired cells)
1.4
1.2
1.0
0.8
0.6
0.4
gate, then items
gate, then items: signal
gate, then items: k-fold noise
90M
175M
350M
600M
1B
1.7B
Model size
Does SNR track DA-size?
Spearman ρ(SNR, DA-size multi-axis)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
361
753
438
494
806
771
403
768
790
546
full benchmark
gate, then items
items, then gate
gate, then items: full task truth
items, then gate: full task truth
90M
175M
350M
600M
1B
Model size